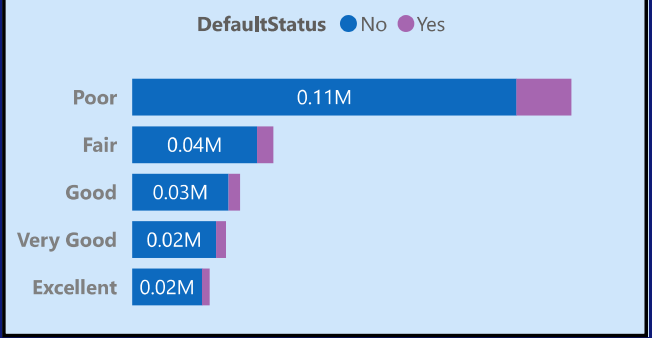
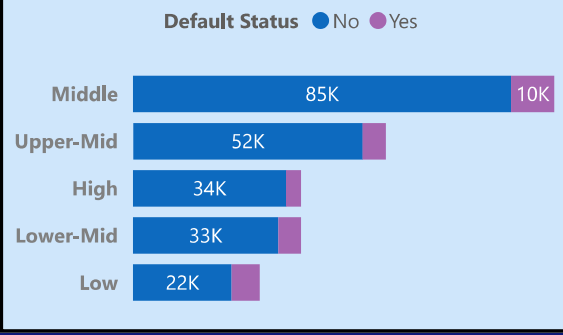
Loan Default Risk Analysis

Total Loans	Total Defaults	Default %	Average Loan Amount	Average Credit Score	High Risk Borrowers (%)
255K	30K	11.6%	127.6K	574.3	54.6%

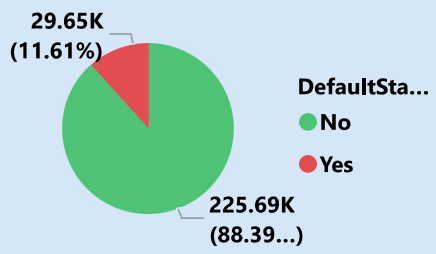
Default Status by Credit Score Group



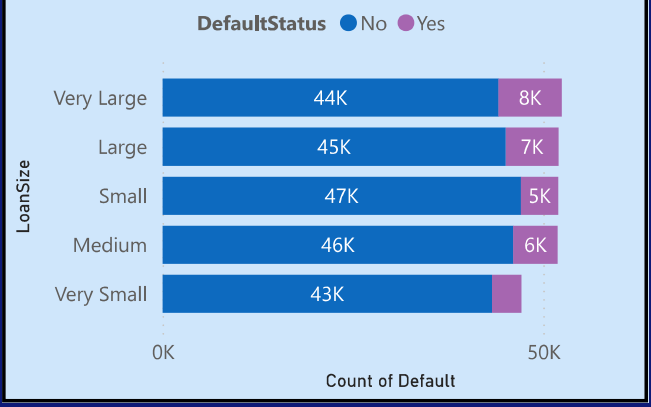
Default Status by IncomeGroup



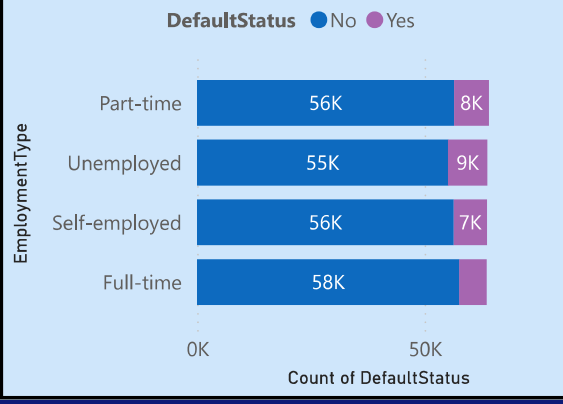
Default Status - Default vs No Defaults



Defaults by Loan Size



Defaults by Employment Type



Mortgage Chart

Mortgage Status	Total Loans	Total Defaul
Yes	127677	138
No	127670	157
Total	**255347**	**296**

Age Groups: All

Credit Lines: All

Marital Status: All

Employment Type: All

Education: All

Loan Default Risk Analysis - Key Insights

Project Overview :

This report analyzes borrower characteristics to identify patterns associated with loan defaults. The analysis was performed using Python for data processing, SQL for data querying, and Power BI for visualization. The objective is to understand the key risk factors influencing loan repayment behavior.

Dataset Summary :

The dataset contains **255,347 loan records** with borrower information including:

- Age
- Income
- Loan Amount
- Credit Score
- Employment Type
- Education Level
- Loan Purpose
- Debt-to-Income Ratio (DTI)
- Default Status

The dataset was cleaned and enhanced using feature engineering techniques to create analytical groups such as **Income Group, Credit Score Group, Age Group, and Loan Size**.

Key Business Questions :

The analysis focused on answering the following questions:

1. What is the overall loan default rate?
2. How does borrower income level influence default risk?
3. What is the relationship between credit score and loan defaults?
4. Which age groups are more likely to default?
5. Does employment type affect default probability?
6. Does loan size influence the likelihood of default?

Loan Default Risk Analysis - Key Insights

Key Insights:

Income Impact

Borrowers in the **low income group show the highest default rate**, indicating that income stability plays a critical role in repayment capability.

Credit Score Risk

Borrowers with **poor and fair credit scores have a higher proportion of loan defaults**, confirming that credit history is a strong predictor of loan risk.

Age Pattern

Younger borrowers in the **18–30 age group show relatively higher default rates**, while older borrowers demonstrate more stable repayment behavior.

Employment Stability

Borrowers with **full-time employment show lower default rates**, suggesting that stable employment significantly reduces credit risk.

Loan Size Effect

Larger loan amounts, particularly **very large loans**, are associated with a higher default probability compared to smaller loans.

Business Recommendations:

Stronger Credit Screening

Financial institutions should apply stricter credit checks for applicants with **low income or poor credit scores**.

Risk-Based Lending Policies

Loan approval criteria can incorporate **credit score and income thresholds** to reduce high-risk lending.

Monitoring High-Value Loans

Larger loan amounts should be monitored closely with enhanced risk evaluation.

Financial Awareness for Young Borrowers

Banks and financial institutions can provide **financial education programs** targeting younger borrowers to improve repayment behavior.

Conclusion

The analysis highlights that **income level, credit score, employment stability, borrower age, and loan size** are key factors influencing loan default risk. By leveraging data-driven insights, financial institutions can improve their **credit risk assessment and lending strategies**.